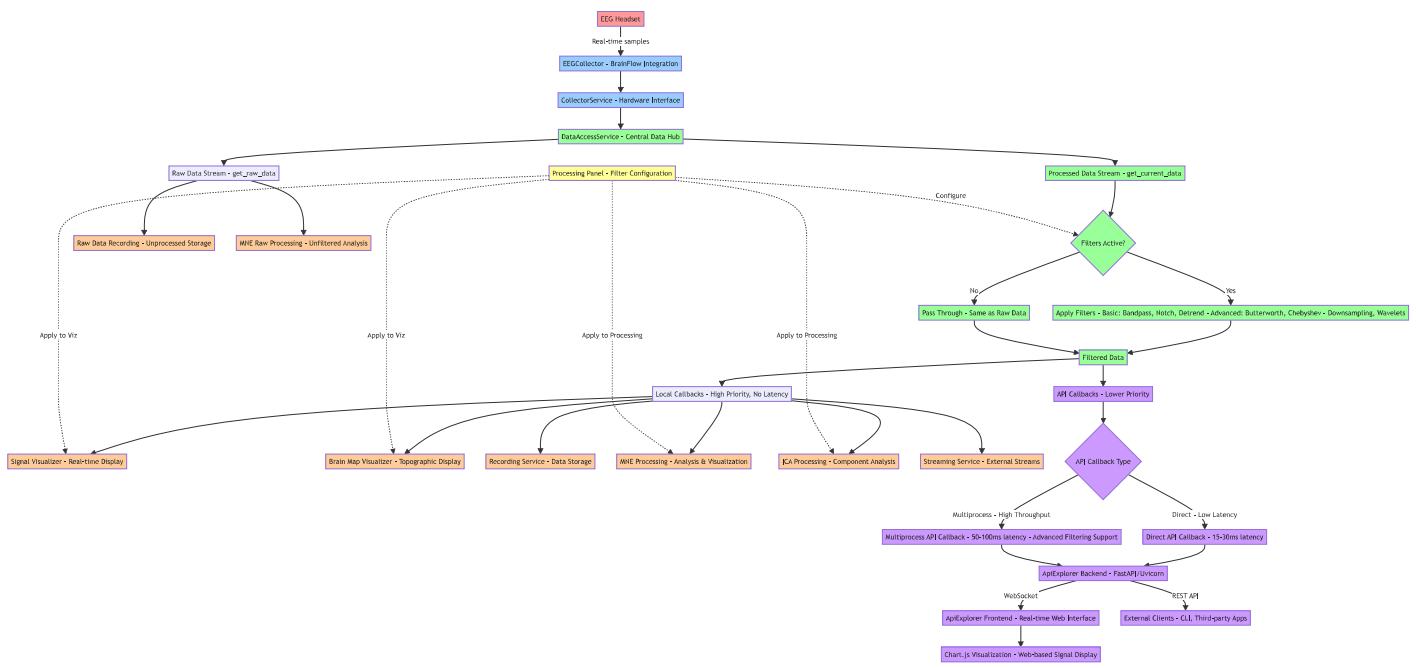

EEG Headset
Real-time samples
EEGCollector - BrainFlow Integration
CollectorService - Hardware Interface
DataAccessService - Central Data Hub
Raw Data Stream - get_raw_data
Processing Panel - Filter Configuration
Processed Data Stream - get_current_data
Configure
Raw Data Recording - Unprocessed Storage
MNE Raw Processing - Unfiltered Analysis
Filters Active?
No
Yes
Pass Through - Same as Raw Data
Apply Filters - Basic: Bandpass, Notch, Detrend - Advanced: Butterworth, Chebyshev - Downsampling, Wavelets
Apply to Viz
Apply to Viz
Apply to Processing
Apply to Processing
Filtered Data
Local Callbacks - High Priority, No Latency
API Callbacks - Lower Priority
Signal Visualizer - Real-time Display
Brain Map Visualizer - Topographic Display
Recording Service - Data Storage
MNE Processing - Analysis & Visualization
ICA Processing - Component Analysis
Streaming Service - External Streams
API Callback Type
Multiprocess - High Throughput
Direct - Low Latency
Multiprocess API Callback - 50-100ms latency - Advanced Filtering Support
Direct API Callback - 15-30ms latency
ApiExplorer Backend - FastAPI/Uvicorn
WebSocket
REST API
ApiExplorer Frontend - Real-time Web Interface
External Clients - CLI, Third-party Apps
Chart.js Visualization - Web-based Signal Display